

When Does Presidential Alignment Buy Federal Grants? Earmark Bans and Executive Discretion

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Abstract

Prior scholarship documents a presidential alignment effect: districts represented by legislators of the president's party receive more federal funds. Comparing discretionary grants before and after Congress banned earmarks in 2011 (110th–111th vs. 112th–116th Congresses), we find that this same-party advantage is institutionally conditional—running through Congress's earmark authority before the ban and through the executive's discretion over agency grants afterward. Under earmarks, the same-party premium spreads broadly across aligned legislators in safe districts rather than concentrating at formal power positions. After the ban, the premium shifts to close-election districts in partisan-safe states—largest in states whose partisan lean runs against the president, smallest in electorally pivotal swing states, the reverse of a swing-vote-targeting logic. The institutional channel thus shapes which districts benefit: legislators target their safe seats, the executive the close-election margin. The political pressure shaping federal grant allocation runs deeper than any single procedural reform.

Keywords: presidential particularism, federal grants, earmarks, distributive politics, executive discretion

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1 Introduction

In the fiscal year 2010 appropriations cycle, members of Congress directed 11,320 specific projects worth \$32 billion to specific recipients in specific districts (Congressional Research Service, 2010). One year later, the FY2011 cycle contained essentially none. The November 2010 election had produced a Republican House majority committed to a self-imposed earmark moratorium; the FY2011 omnibus failed in the lame-duck Senate; the 112th Congress passed a full-year continuing resolution with no earmarks. By FY2012, the appropriations process under the new moratorium contained zero. The federal dollars that had previously flowed through legislator-directed earmarks did not disappear. They passed instead through executive-discretionary allocation channels.

This institutional shift created an unusual opportunity to test distributive politics. If presidentially aligned districts received more federal funds because legislators in the president's party used earmarks to direct dollars home, the alignment effect should weaken or disappear once earmarks were banned. If, instead, alignment effects emerge from executive discretion in distributing federal grants, the post-2011 era should reveal them, perhaps even amplified, since the earmark channel that previously absorbed legislator-directed targeting was no longer available. The two predictions diverge sharply: same dollars, same partisan motives, different institutional channel.

We ask two related questions. First, does presidential alignment causally increase a district's federal discretionary grants? Second, does the magnitude of this alignment effect depend on whether legislators retain direct earmark authority? To answer the first credibly, we exploit close House elections: where the president's party narrowly won or lost a seat, alignment is as good as randomly assigned. We answer the second by estimating the same close-election regression discontinuity, on the same federal grant categories, separately before and after the 2011 earmark ban.

The institutional logic of partisan grant targeting operates through two channels. Legislators with direct earmark authority can write specific projects into appropriations

bills, ensuring that federal dollars flow to specific recipients in specific districts without further executive intervention. Executive officials, in turn, exercise discretion over the much larger set of competitive discretionary grants. Both channels can in principle carry partisan-alignment effects. Legislators in the president's party may use earmarks to deliver dollars to constituencies that support the administration; executive agencies may favor districts represented by aligned members in discretionary award decisions. The relative importance of these two channels depends on which is institutionally available.

This substitution structure produces a sharp prediction. When earmarks are institutionally available, members of Congress across both parties direct federal dollars to their districts through earmarks; at the close-election margin in particular, we should observe no detectable targeting advantage for legislators of the president's party. The legislator channel does generate alignment effects in the broader district population, broadly distributed among aligned legislators in structurally safe districts and across the chamber rather than channeled through formal power positions; but because this is a safe-seat phenomenon, the effect relocates away from the competitive margin, leaving little signal there and remaining largely invisible to a close-election regression discontinuity. Within the executive channel, presidents have political incentives to favor aligned districts in the discretionary grant decisions agencies make, but when earmarks have already pre-allocated a substantial portion of the relevant grant budget, the executive's room to act on those incentives is constrained. Once earmarks are banned, the legislative channel that previously absorbed targeting decisions is closed. The executive channel becomes the operative vehicle for whatever partisan-alignment-driven allocation occurs. Aligned districts should observably benefit at the close-election margin only after the ban, the empirical signature of substitution.

The argument has three commitments worth making explicit. The first is an institutional reading of distributive politics: alignment effects are not stable preferences expressed through fixed channels but outcomes shaped by which procedural tools are avail-

able. The second is a focus on discretionary grants—FABS award types 04 (Project Grant) and 05 (Cooperative Agreement)—rather than aggregate federal spending; this category is the one most amenable to political targeting and is also the category most directly affected by earmark substitution. The third is an identification choice rather than a substantive one: we restrict the cleanest causal comparison to the close-election margin, where the running variable produces quasi-random treatment assignment, and complement it with panel methods for the safe-seat margin where the legislator channel operates. We test the prediction by applying the same close-election regression discontinuity design separately within the pre- and post-ban regimes.

We exploit the close-election regression discontinuity design at the boundary where the president’s party narrowly won or lost a House seat. The running variable is the difference between the president’s-party share of the two-party House vote and 50 percent; treatment is defined as the president’s party winning the seat (running variable above zero). The outcome is the log of discretionary federal grants—FABS award types 04 and 05—summed over fiscal years $t+2$ and $t+3$, the years whose appropriations the new legislator’s chamber actually voted on. We use this window rather than the more conventional $[t+1, t+2]$ window because FY $t+1$ appropriations are written by the *outgoing* Congress in fall of year t , before the new legislator is even sworn in.

Our sample comprises six House election cycles spanning the two regimes. The pre-2011 Traditional regime is represented by the 2006 and 2008 elections, whose corresponding 110th and 111th Congresses wrote appropriations under the traditional earmark process. The post-2011 ban regime is represented by the 2010, 2014, 2016, and 2018 elections, whose 112th through 116th Congresses wrote appropriations entirely under the moratorium. Election-cycle observations are dropped at redistricting boundaries to avoid attribution problems. To test whether the pre-ban point estimate is an artifact of the two-cycle FABS sample, we extend the pre-DATA-Act sample by integrating Census Bureau Consolidated Federal Funds Reports with USASpending Federal Award and Subaward

Subsystem data, validating the integration on overlapping post-2007 fiscal years.

We complement the close-election RDD with two heterogeneity strategies. A panel-fixed-effects analysis stratified by lagged district margin tests whether the alignment effect under the legislator channel is detectable in structurally safe seats and absent at the close-election margin. A Banned-regime RDD stratified by state-level presidential margin—further split by whether the state’s partisan tilt aligns with or opposes the sitting president—tests whether the executive channel follows a swing-vote persuasion logic in which targeting concentrates in electorally pivotal states, or instead a Bayesian electoral-information logic in which targeting concentrates where close-election outcomes are most informative given state-level priors. Each strategy probes a distinct channel signature.

The empirical pattern is consistent with institutional substitution. Under the post-2011 ban regime, presidentially aligned districts receive roughly two-thirds more federal discretionary grants at close-election margins than misaligned districts; the effect approaches a doubling when the transitional 2010 election cycle is excluded. The same regression discontinuity design applied to the pre-ban Traditional regime yields no detectable effect; the Traditional point estimate stays near zero when the sample is doubled with Census Bureau pre-DATA-Act records, though the Traditional design is underpowered to rule out an effect the size of the post-ban estimate, so we read this null as consistent with rather than confirmation of the substitution prediction. The post-ban effect is concentrated in fiscal years following the new legislator’s installation and not detectable in pure pre-election placebo periods. A first-difference robustness specification preserves the regime contrast.

The two channels leave distinct empirical signatures across two different empirical lenses, each at the location predicted by its respective channel. Heterogeneous panel-fixed-effects analysis under the Traditional regime locates a positive alignment premium of roughly nineteen percent within the safe-district stratum, distributed broadly across aligned safe-seat legislators rather than concentrated at formal power positions, and invisible at the close-election margin where institutional positions do not vary systemati-

cally with alignment. Under the post-ban regime, the close-election RDD effect is driven entirely by partisan-safe states and is absent in swing states, running opposite to Kriner and Reeves (2015)’s electoral-particularism prediction that presidential targeting concentrates at the electorally pivotal swing-state margin; within the safe stratum the effect is larger in opposition strongholds (where a co-partisan close-election win is most surprising) than in co-partisan strongholds, the asymmetric signature predicted by a Bayesian electoral-information logic. The pre-ban legislator channel and the post-ban executive channel produce two distinct empirical signatures that, taken together, trace the substitution mechanism through both directions. The formal regime-interaction test we report (Section 4.6) is borderline at conventional thresholds; we therefore rest our central claim on the joint pattern these signatures trace, not on that single formal contrast.

Three contributions follow.

First, we provide what is, to our knowledge, the cleanest causal identification of presidential alignment effects on federal grant allocation in the post-DATA-Act era. Panel-fixed-effects estimators on aggregate federal-spending categories have documented alignment effects in this prior empirical literature (Berry et al., 2010; Albouy, 2013; Larcinese et al., 2006). These methods generate informative average effects but pool across districts of all electoral competitiveness and across institutional regimes. Dynes and Huber (2015) sharpen the question of who benefits—same-party legislators or same-party voters—using within-district designs that exploit changes in institutional control while the seat’s occupant stays constant, and they note the sorting concerns that close-election designs face in House data; we adopt the close-election margin precisely because its identifying assumptions are directly testable, verify them in our sample (Section 3), and extend the design to the post-DATA-Act discretionary-grant outcome and, centrally, to the institutional-regime contrast. Close-election regression discontinuity restricts identification to the pivotal close-election margin where targeting decisions are most plausibly contested, and where the underlying partisan composition is quasi-randomly assigned.

On the narrower discretionary-grant outcome, the post-ban estimate we recover substantially exceeds the panel-FE benchmarks reported in prior work, consistent with either close-election winner's-curse inflation or with a genuine LATE-versus-ATT gap at the contested margin.

Second, and substantively most novel, we find that the alignment effect is institutionally conditional and that the two channels leave distinct empirical signatures. Prior literature treats the alignment effect as a relatively stable feature of executive influence over distributive politics; our regime contrast indicates that the effect is closer to an institutionally produced outcome than a baseline preference. Under the Traditional earmark regime, the same close-election design on the same outcome finds no detectable alignment effect; under the post-ban regime, the effect emerges and is robust across bandwidth, polynomial, and first-difference specifications. The pattern is consistent with a substitution mechanism between legislative and executive distributive channels—a logic whose building blocks are long-standing in the policy-subsystems account of a federal pork barrel that flows through agency-administered grant programs and adapts to institutional reform (Stein and Bickers, 1995; Bickers and Stein, 1996)—but it has not previously been tested with a sharp institutional shift. The 2011 ban is, in this sense, the first natural experiment that lets us isolate one channel from the other within a single empirical design. Beyond the regime contrast, the two channels concentrate the alignment effect at different features of the district population: under earmarks, alignment effects obtain within structurally safe seats and distribute broadly across aligned legislators rather than concentrating at formal power positions; under the post-ban executive channel, alignment effects concentrate at the close-election margin within partisan-safe states rather than in electorally pivotal swing states, and within the safe stratum are larger in opposition strongholds than in co-partisan strongholds. The second pattern refines the swing-state-targeting prediction implied by Kriner and Reeves (2015)'s electoral-particularism channel; their separate partisan-particularism channel (core-state targeting) is partially

consistent with our positive safe-co-partisan estimate, and their coalitional-particularism channel is engaged by our heterogeneous-TWFE legislator-channel results. Within the scope where executive discretion is most consequential for grant allocation, the empirical pattern points toward an information-/scarcity-based logic—developed below as a Bayesian electoral-information account—in which surprising co-partisan close-election wins draw the largest grant response, with swing-vote persuasion playing no detectable role at the within-state across-district margin.

Third, we contribute to the methodological infrastructure of distributive politics research by integrating Census Bureau Consolidated Federal Funds Reports (CFFR) with USASpending FABS data to extend district-level grant histories back to the early 1990s. The pre-DATA-Act period has been a coverage gap for transaction-level grant analysis at the congressional district level: FABS records pre-FY2007 carry congressional district codes on roughly 0.4 percent of project-grant transactions. CFFR reports, prepared annually by the Census Bureau from 1983 through 2010, fill this gap with congressional-district-level dollar totals at coverage rates above 50 percent. We document the integration’s validity on overlapping post-2007 fiscal years and use it to show that the Traditional-regime point estimate remains near zero when the historical sample is doubled—a location-stability check, though the extension does not materially sharpen the null’s precision. The merged panel will be released as a public-use dataset.

2 Theory and Institutional Context

2.1 Two Channels of Partisan Grant Targeting

Federal grant allocation flows through two distinct institutional channels, each with its own mechanism for partisan targeting. The legislative channel operates through appropriations: members of Congress write specific projects, programs, or recipients directly into law, securing dollars for their districts before agency discretion enters the picture.

Earmarks have historically been the most direct form of legislative-channel targeting, but the channel also operates through committee report language and informal lobbying during conference negotiations. When Congress directs a specific dollar amount to a specific recipient—in bill text or in the committee-report language that agencies treat as binding statements of congressional intent—agency officials have little room to redirect those dollars toward the executive’s preferences (Frisch and Kelly, 2011; Evans, 2004).

The executive channel operates through agency discretion within already-passed appropriations. Most federal grant programs include broad statutory authority for agencies to allocate dollars across competitive applications based on agency-defined criteria. Within these decisions, political pressure shapes outcomes: agency officials respond to White House priorities, presidential appointees set program emphases, and elected officials influence individual award decisions through informal communication and constituency casework. Empirical scholarship has documented an aggregate alignment effect on federal spending — districts represented by members of the president’s party receive systematically more federal funds than districts represented by the opposition (Berry et al., 2010; Albouy, 2013), with panel-fixed-effects coefficients on aggregate spending in the range of five to ten percent; executive-branch allocation studies show that department-level grant decisions likewise respond to political considerations (Bertelli and Grose, 2009). These aggregate-spending estimators do not, on their own, attribute the alignment effect to either channel: legislative-channel and executive-channel flows are pooled in the dependent variable.

These two channels are not independent. They draw from the same finite federal grant budgets. Dollars committed to a member’s district through earmarks are pre-allocated by statute and cannot be redirected by agency discretion. When earmarks are widely available, the legislative channel absorbs a substantial portion of the targeting decisions for a given fiscal year’s grant budget; the executive channel operates on the residual. When earmarks are unavailable, agencies retain control over the full discretionary budget, and

the political pressures shaping that allocation become more consequential. The relative importance of these two channels for partisan targeting is not a fixed feature of the federal system. It depends on which procedural tools legislators have at their disposal.

2.2 Institutional Context: Earmarks and the 2011 Ban

Earmarks were a routine, high-volume feature of federal appropriations from the 1980s through 2010. The FY2010 appropriations cycle contained 11,320 earmarks worth \$32 billion (Congressional Research Service, 2010), with comparable volumes in immediately preceding years. Members of Congress requested earmarks through their committees of jurisdiction; appropriations subcommittee leadership negotiated the final allocation; the resulting projects and dollar amounts appeared in the bill text or accompanying committee reports. The institutional process was bipartisan: both Democrats and Republicans in both chambers participated, and successive internal reforms had by the 2000s extended earmark access to virtually all members of both parties and both chambers, even as appropriators and the majority party continued to claim disproportionate shares (Frisch and Kelly, 2011; Lazarus and Steigerwalt, 2009).

The moratorium emerged from a political confluence in late 2010. The November 2010 election produced a Republican House majority committed to an earmark ban; the Senate Republican Conference adopted a parallel moratorium in early 2011; President Obama announced he would veto any bill containing earmarks. The lame-duck 111th Congress attempted to pass an FY2011 omnibus containing roughly \$8.3 billion in earmarks, but the bill failed in the Senate after Republican opposition crystallized. Congress instead enacted a series of continuing resolutions, culminating in a full-year continuing resolution (P.L. 112-10) in April 2011 with no earmarks. The 112th Congress wrote the FY2012 appropriations cycle entirely under the formal moratorium. The ban remained in place through the 116th Congress (2019–2021).

The 117th Congress (2021–2023) partially restored legislator-directed funding under

new procedures: the House renamed the practice Community Project Funding and the Senate renamed it Congressionally Directed Spending, each with per-member caps, transparency requirements, and prohibitions on for-profit recipients. The post-2022 restored regime falls outside our main analysis window for two reasons. First, only the 2020 election cycle is fully observed under the appropriations-writing window we use; the 2024 cycle’s spending years extend beyond available USASpending data. Second, the 2020 cycle’s pre-period placebo fails: the running variable is correlated with pre-existing district-level grant differentials that emerged in the COVID era, preventing a clean reading of the alignment effect under the restoration. We return to this restoration in the conclusion as a future research opportunity.

2.3 The Substitution Argument

The argument proceeds in three steps.

Under the traditional appropriations process, members of Congress across both parties directed federal dollars to their districts through earmarks. The earmark process was bipartisan, and the advantages it conferred tracked congressional position—majority status, committee membership, seniority—rather than shared partisanship with the president; there is no consistent evidence that members of the president’s party received systematically more earmark dollars than members of the opposition (Bickers and Stein, 2000; Levitt and Snyder, 1995; Lazarus and Steigerwalt, 2009). Because earmark advantages track position rather than alignment, presidential alignment confers no earmark-targeting advantage at the close-election margin: aligned and misaligned legislators alike use earmarks to deliver dollars to their districts, and the average earmark-driven flow is roughly equal across the close-election cutoff.

Within the executive channel during the same era, presidents and their agencies have political incentives to favor aligned districts in discretionary grant decisions, but those incentives operate on a constrained budget after earmark pre-allocation. The empirical im-

plication is that the alignment effect captured by a close-election regression discontinuity on FABS discretionary grants should be muted in the pre-ban era: the residual executive room is too small to produce a detectable coefficient at the close-election margin.

Once the moratorium takes effect, the legislative channel closes. Members of Congress can no longer pre-allocate specific dollars to specific projects in their districts. The full discretionary grant budget—formerly partly committed via earmarks—becomes subject to agency discretion within statutory authority. Presidents and their agencies retain the same incentives to favor aligned districts, but they now have the institutional room to act on those incentives. Aligned districts should receive systematically more discretionary grants, with the magnitude of the effect reflecting the size of the previously-earmarked budget that has been freed for executive allocation. Our close-election regression discontinuity on FABS discretionary grants in the post-ban era should detect this effect.

The two channels do not produce identical empirical distributions when they operate. Under the traditional regime, the legislator channel routes federal dollars through the earmark process. Earmark access by the 2000s extended to virtually all members of both parties (Frisch and Kelly, 2011), and within this broad participation the alignment effect on discretionary grants obtains within structurally safe districts—a pattern we document empirically in Section 4.2. The legislator channel therefore produces an alignment effect that obtains among aligned legislators in structurally safe districts, leaving little detectable signal at the close-election margin: the effect is broadly distributed across aligned safe-seat legislators rather than channeled through institutional position, so the competitive margin—which by construction excludes safe seats—reveals none of it. This safe-stratum signature is invisible to a close-election regression discontinuity but detectable in heterogeneous panel methods stratified by district safety.

The executive channel, by contrast, can operate across all margins within the discretionary budget. Under the post-ban regime, when agencies and the White House can redirect dollars across districts on whatever political logic motivates them, the alignment

effect concentrates at the close-election margin: aligned-winning districts barely held by the president's party are politically valuable to retain, and the executive has both the incentive and the institutional room to direct grants accordingly. The two regimes therefore produce two distinct empirical signatures—a legislator-channel safe-seat signature in the Traditional era, an executive-channel close-election signature in the Banned era—and a comprehensive test of the substitution mechanism examines both signatures, not the regime contrast alone.

Three alternative mechanisms warrant consideration. First, the 2009 ARRA injected an unusually large volume of executive-discretionary grants into the early ban era; ARRA could plausibly be one operative manifestation of the substitution mechanism rather than a confound. Excluding the 2010 cycle, which captures the largest ARRA concentration, strengthens the post-ban estimate. Second, pre-existing partisan sorting at the close-election margin could generate the appearance of an alignment effect; we address this through a fiscal-year-level placebo decomposition and a first-difference robustness specification. Third, the regime contrast could reflect identification quality differences rather than an institutional shift; we address this by extending the pre-ban sample with Census Bureau pre-DATA-Act records, doubling the Traditional-regime observations, and the pre-ban null persists across both data sources.

The substitution argument generates a testable implication that the remainder of the paper examines. The same close-election regression discontinuity design, applied to the same outcome, within the same fiscal-year window, but separately within the Traditional and Banned earmark regimes, should yield a regime-conditional alignment effect: null in the pre-ban era and positive in the post-ban era. The contrast between the two regime estimates is the empirical signature we expect under the institutional substitution mechanism.

2.4 Two Logics of Presidential Particularism: Swing-Vote Persuasion versus Bayesian Electoral Information

The literature on distributive politics has long recognized that presidents engage in strategic targeting of federal grants, but it has not converged on which locations they prioritize. Kriner and Reeves (2015) formalize a three-channel particularism framework (hereafter K&R): presidents target electorally pivotal *swing states* (electoral particularism), *core states* where the president's party averages disproportionately strong electoral support (partisan particularism), and districts represented by *co-partisan legislators* in Congress (coalitional particularism, the alignment-effect mechanism documented by Berry et al. 2010). All three channels predict positive spending at their respective targets, with hostile states—those leaning against the president's party—as the omitted baseline. Of these channels, electoral particularism generates the sharpest prediction about *alignment-effect heterogeneity* at the close-election margin: it predicts that the close-election alignment premium should concentrate in states where the next presidential election is competitive. Empirical work in this tradition documents precisely such swing-state targeting in the allocation of discretionary federal grants (Hudak, 2014). We label this strand the *swing-vote persuasion logic*, anchored in the formal redistributive-politics tradition of Lindbeck and Weibull (1987) and Dixit and Londregan (1996) in which parties target persuadable voters in electorally pivotal districts and underweight strongholds whose voters are already committed. Aligned legislators in partisan strongholds, on this view, are not the priority; their voters are already locked in. The opposing formal tradition—the core-voter model of Cox and McCubbins (1986), in which risk-averse parties invest first and foremost in their loyal support groups and treat swing groups as riskier investments—predicts the reverse allocation, and it shares with the Bayesian electoral-information logic we develop below the expectation that targeted resources flow to safe rather than swing constituencies.

We develop an alternative logic for the post-ban executive channel that we call the *Bayesian electoral-information logic*. Under this account, the political return on a discre-

tionary grant scales with the informational value of the close-election outcome that produced the legislator. State-level partisanship sets a prior over which party is expected to hold any given district. A close-election outcome that defies the state-level prior—a co-partisan winning narrowly in opposition territory, or a co-partisan losing narrowly in friendly territory—is highly informative; an outcome that confirms the prior is not. The executive’s grant lever, however, is asymmetric: it can reward a co-partisan winner but cannot directly reward an opposition winner. Combining the informational and asymmetric-lever components yields cross-stratum predictions on the magnitude of the close-election RDD coefficient.

To formalize, let s index a state-level prior over the president’s-party share, and let $|s - 0.5|$ measure state partisan safety. The just-above-cutoff side of the RDD captures co-partisan close wins; the just-below side captures opposition close wins. Under the swing-vote persuasion logic, the close-election RDD coefficient $\tau(s)$ should peak in swing states where $|s - 0.5|$ is small, because dollar-per-vote returns are highest where voters are persuadable. Under the Bayesian electoral-information logic, $\tau(s)$ depends on whether the just-above outcome (the co-partisan win) was surprising given the state prior. In opposition strongholds, where s favors the opposition, a co-partisan close win is highly surprising, so the just-above side draws the largest grant response and τ is largest. In co-partisan strongholds, where s favors the president’s party, a co-partisan close win is expected, so the just-above response is moderate and τ is smaller; the highly surprising outcome on the just-below side—a co-partisan losing in friendly territory—cannot be rewarded because the legislator who took the seat is opposition. In swing states, neither just-above nor just-below outcomes are particularly informative, so τ is near zero. The Bayesian-information logic therefore predicts an asymmetric U-shape across the safety gradient, with $\tau_{\text{safe-opposition}} > \tau_{\text{safe-co-partisan}} > \tau_{\text{swing}} \approx 0$.

The two logics produce empirically distinguishable predictions at the within-state across-district margin. Swing-vote persuasion predicts the close-election RDD coefficient

peaks in swing states. Bayesian electoral information predicts the coefficient peaks in opposition strongholds, with co-partisan strongholds intermediate and swing states near zero. The interaction structure between alignment, state partisan safety, and the sign of the state’s partisan tilt relative to the sitting president distinguishes the two logics directly.

This contrast applies most directly to the post-ban executive channel, where the executive is the primary decision-maker. Under the legislator channel, the contrast may operate as well, either through aligned legislators in partisan-safe states coordinating with their party’s president on agency execution, or through residual executive discretion operating on the constrained budget that earmarks leave behind. The contrast becomes most empirically discernible, however, in the post-ban regime, when executive discretion is the primary route for partisan targeting. Standard panel-fixed-effects estimators in the K&R tradition aggregate across both swing and safe states; they cannot adjudicate between the two logics on their own. The close-election regression discontinuity in our design, stratified by state-level swingness and further by sign relative to the president’s party, produces the necessary contrast.

We do not view the test we develop as a rejection of K&R’s broader framework. K&R’s coalitional particularism and their partisan particularism (core-state targeting) are not contradicted by our findings. The coalitional channel is engaged separately by our heterogeneous-TWFE legislator-channel results (Section 4.2). The partisan channel is partially consistent with our safe-co-partisan stratum estimate ($\tau = +0.889$ in the Banned regime; positive but noisier). What our findings sharply refine is K&R’s *electoral particularism*: the swing-vote-persuasion prediction is directly contradicted by our $\tau_{\text{swing}} \approx 0$ paired with $\tau_{\text{safe-opposition}} > 0$. Two scope qualifications apply throughout. K&R’s claims are about state-level *level* differences in spending averages, identified by panel fixed-effects across both swing and safe districts; ours is about the LATE at the close-election margin, where partisan composition is locally randomized. K&R use total federal grants including formula channels; we restrict to the discretionary subset where executive discretion is

least constrained. Within this narrower scope—discretionary grants at the close-election margin in the post-ban era—the empirical pattern points toward a Bayesian electoral-information logic, with surprising co-partisan close-election wins drawing the largest grant response and electorally pivotal swing states drawing none.

3 Research Design

3.1 Data and Outcome

Federal grant transactions come from the USASpending Financial Assistance Broker Submission (FABS) for FY2007 onward and the Census Bureau Consolidated Federal Funds Reports (CFFR) for FY1993–2010. We classify federal grant types by FABS `assistance_type` and restrict to type 04 (Project Grants) and type 05 (Cooperative Agreements), which together comprise the discretionary-grant category most amenable to executive targeting. Block grants and formula grants are excluded because their allocation rules constrain executive discretion to within statutory formulas; direct payments to individuals are excluded as not relevant to the partisan-targeting question.

The outcome variable is the natural logarithm of `discretionary_assistance` plus one dollar, summed across two fiscal years. The window choice is motivated by the appropriations-writing channel: appropriations for fiscal year $t+1$ are written in late summer of year t before the November election, whereas appropriations for fiscal year $t+2$ onward fall fully under the new legislator’s term. We therefore use the window $[t+2, t+3]$ as the primary outcome and report $[t+1, t+2]$ as a robustness check that captures the agency-execution channel during the hybrid first fiscal year.

District-fiscal-year observations are constructed by aggregating individual transactions to congressional district within fiscal year. Transactions with missing or unparseable district codes are dropped; transactions identified to multi-district recipients are not allocated across districts. CFFR pre-2007 data are integrated into a merged spending panel;

we document the integration’s behavior on overlapping FY2007 in two ways. First, FABS pre-FY2008 carries congressional district codes on roughly 0.4 percent of project-grant transactions, so direct CD-level FABS aggregates for pre-2007 fiscal years are unusable; CFFR’s county-to-CD allocation algorithm recovers approximately 99 percent of CFDA-04/05 grants to specific districts. Second, on overlapping FY2007 (where FABS coverage begins), the two sources agree at the district-level Spearman rank correlation of 0.62 in levels and Pearson 0.59 in logs, with the CFFR aggregate exceeding the FABS aggregate by roughly 30 percent because CFFR captures grants that FABS lost to blank-CD coding before FY2008.¹ The level discrepancy is large; what matters for our identification is whether the *RDD coefficient* is invariant across the two data sources within the Traditional regime. Section 4.1 and Section 4.6 establish that it is: the Traditional FABS-only and CFFR-extended Traditional RDD coefficients are nearly identical (Table 1, columns 1 and 3) despite the level-coverage gap.

3.2 Scope of the Discretionary-Grant Outcome

The discretionary-grant scope is narrower than the federal-spending categories used in prior work, but the right denominator for assessing whether the scope is substantively meaningful is not the total federal budget. Mandatory programs (Social Security, Medicare, Medicaid, statutory entitlements) constitute around 60 percent of annual federal outlays across our sample, which averaged approximately \$4 trillion per year and rose above \$6 trillion during the COVID period (Congressional Budget Office, 2026), and the largest grant-to-states program—the Medicaid match—is itself a statutory formula; both lie outside any plausible scope for executive partisan targeting. FABS assistance types

¹Validation diagnostic: 437 districts in the merge intersection on FY2007; CFFR aggregate \$19.0B, FABS aggregate \$13.0B; mean absolute district-level discrepancy of 64 percent with median 54 percent reflecting the persistent FABS pre-2008 coding gap rather than substantive disagreement about grant flows. The transition to FY2008+ FABS coverage immediately stabilizes the source-by-source agreement; our CFFR-extension is used only for the Traditional regime sample doubling, where data-source robustness depends on RDD-coefficient invariance, not on level agreement.

04 and 05—the discretionary pool we measure—average approximately \$90 billion per year in normal years across the sample window, rising above \$200 billion during the ARRA (FY2009–2010) and COVID (FY2020–2021) periods: a small share of total federal outlays (approximately 2.3 percent in normal years), but essentially the entire pool over which agency officials retain meaningful allocation discretion. The alternative discretionary channel, FAR-governed procurement, we examine separately and find no analogous alignment effect (Section 4.6); that null bounds the alignment phenomenon to the grant pool rather than to executive discretion writ large. The 2010 earmark inventory of \$32 billion (Congressional Research Service, 2010) is roughly one-third of the post-2007 04+05 pool—small relative to the federal budget but a non-trivial fraction of the slice that would be freed for executive reallocation under the moratorium, and large enough that a substitution effect on close-election margins is mechanically plausible. Our findings accordingly concern the discretionary-grant slice where executive discretion is operative and cannot speak to the much larger mandatory or formula-grant categories where allocation is constrained by statute.

3.3 Sample and Regime Definition

The unit of observation is a district-election (state, congressional district, election year). House election results since 2000 come from the MIT Election Lab data; we drop election years immediately following decennial redistricting (2002, 2012, 2022) because district boundaries shift and the running variable is not comparable to prior cycles within the same district. Our analytic sample comprises the 50 states; the District of Columbia is excluded structurally because it has no voting House seat.

We define two earmark regimes by the Congress that wrote the appropriations governing each outcome window. The Traditional regime spans the 110th and 111th Congresses (FY2007–FY2010), during which earmarks were a routine, high-volume feature of appropriations. The Banned regime spans the 112th through 116th Congresses (FY2011–

FY2021), during which the earmark moratorium was in formal effect across both chambers. The 117th Congress’s partial restoration (FY2022 onward) falls outside our main sample.

Election years assigned to each regime under the $[t+2, t+3]$ window: Traditional = {2006, 2008} and Banned = {2010, 2014, 2016, 2018}. Both windows assign election year t cleanly to a single regime when the appropriations-writing Congress is unambiguous; the $[t+1, t+2]$ alternative assigns 2008-election spending across the Traditional–Banned boundary because FY2011 is the regime-transition fiscal year. We report results separately for both window definitions.

3.4 Identification: Close-Election RDD

Identification rests on the close-election regression discontinuity design first developed for U.S. House races by Lee (2008). The running variable is `margin_pres_party`, defined as the House two-party vote share for the president’s party minus fifty percent; the treatment indicator equals one when the president’s party wins the House seat. We estimate the local linear regression discontinuity using the robust bias-corrected RD estimator of Calonico et al. (2014) with MSE-optimal bandwidth selection; our implementation uses a triangular kernel and clusters standard errors at the state level to account for within-state correlation across districts.

The RDD identifies the local average treatment effect at the close-election margin. This margin is the empirical signature predicted under the executive channel: a barely-aligned district just past the cutoff is, on the substitution argument, the location where presidential discretion has both the political incentive and the institutional room to operate. The same design at the same margin in the Traditional regime is predicted to yield a null estimate, because the legislator channel operates among aligned legislators in structurally safe districts rather than at the close-election margin.

The identifying assumption is local random assignment of treatment status at the cut-

off. Close U.S. House races are the canonical setting in which sorting at the cutoff has been documented (Caughey and Sekhon, 2011), though broader evidence across electoral settings finds such sorting to be the exception rather than the rule (Eggers et al., 2015); we therefore test the assumption directly in our sample. Two pieces of evidence support local random assignment. We test for manipulation of the running variable using the local-polynomial density discontinuity procedure of Cattaneo et al. (2018), which extends the original density test in McCrary (2008). No significant manipulation is detected around the cutoff in either regime. Pre-treatment covariate balance tests at the discontinuity show no significant difference between aligned-winning and aligned-losing districts on prior-cycle margin or earlier-period grant levels in the Traditional regime; in the Banned regime, the prior-cycle margin discontinuity is borderline (consistent with mild persistence of partisan margins across cycles), and the placebo decomposition in Section 4.5 directly addresses any residual concern through the fiscal-year structure of pre-period estimates. Full identification-check results appear in Appendix A.

3.5 Heterogeneous TWFE Specification

The close-election RDD captures the close-election margin only. To detect the legislator channel—predicted to operate among aligned legislators in structurally safe districts—and to adjudicate the swing-vote-persuasion versus Bayesian-electoral-information contrast at the state level, we use a complementary heterogeneous panel-fixed-effects design.

For each district-election we construct two safety measures. The first, district safety, is the absolute prior-cycle `margin_pres_party` in the same district. Districts are classified as competitive ($|\text{lag margin}| < 5\text{pt}$), moderate (5–10pt), or safe ($>10\text{pt}$). The second, state presidential swingness, is the signed Republican two-party margin in the state’s most recent presidential election (2004, 2008, 2012, or 2016). States are classified by absolute margin as swing ($|m| < 5\text{pt}$), lean (5–10pt), or safe ($>10\text{pt}$); the lean and safe categories are further split by whether the state’s partisan tilt aligns with (*co-partisan*) or

opposes (*opposition*) the sitting president’s party at each election cycle. The first measure operationalizes the legislator-channel safe-seat hypothesis from the prior subsection; the second operationalizes the swing-vote-persuasion versus Bayesian-electoral-information contrast.

The heterogeneous specifications regress the log discretionary grants outcome on alignment, safety category, and their interaction, with state and year fixed effects in the baseline specification and district and year fixed effects in the stricter alternative. Standard errors are clustered at the state level in the baseline and at the state-by-district level in the district-FE specification. The interaction coefficient on alignment $\times$ safety reveals which type of district receives the alignment premium under each regime; we report joint tests of the conditional alignment effect within each safety stratum (the linear combination of the main alignment coefficient and the stratum interaction) since the conditional effect is the policy-relevant quantity rather than the interaction coefficient alone.

A recent literature has documented systematic concerns with two-way fixed-effects estimators when treatment effects are heterogeneous across units or over time (Goodman-Bacon, 2021; de Chaisemartin and D’Haultfœuille, 2020; Sun and Abraham, 2021; Borusyak et al., 2024). The leading concern — negative weighting of already-treated comparison observations — arises principally in staggered-adoption settings where a single TWFE coefficient implicitly averages effects across cohorts. Our specifications sidestep this concern in two ways. First, we report stratum-conditional joint tests rather than a single TWFE coefficient, so each reported quantity is a within-stratum local effect rather than a cross-stratum weighted average. Second, alignment is not a staggered absorbing treatment; districts move in and out of aligned status election by election, so the imputation-estimator critique of Borusyak et al. (2024) has limited bite for our setting. We retain TWFE rather than switching to a recent alternative estimator because the design’s identification rests on within-stratum comparisons that the canonical TWFE

delivers transparently in our non-staggered panel.

Four predictions follow from the theory section. Under the Traditional regime with district-safety stratification, the alignment effect should be positive in safe districts and smaller or absent at the close-election margin (legislator-channel signature). Under the Traditional regime with state-pres-swing stratification, the swing-vote-persuasion logic predicts a peak effect in swing states while the Bayesian-electoral-information logic predicts the peak within partisan-safe states with the largest effect in opposition strongholds. Under the Banned regime with district-safety stratification, the alignment effect should be flat or weakly positive across strata because the legislator channel is closed. Under the Banned regime with state-pres-swing stratification, the swing-vote-persuasion versus Bayesian-electoral-information contrast bites: swing-vote persuasion predicts swing-state concentration, while Bayesian electoral information predicts the asymmetric ordering $\tau_{\text{safe-opposition}} > \tau_{\text{safe-co-partisan}} > \tau_{\text{swing}} \approx 0$ at the close-election margin.

3.6 Identification Threats and Absorption

The two identification strategies absorb different confounders. The district fixed effects in the heterogeneous TWFE specification absorb every district-time-invariant trait, including poverty rate, urban–rural composition, baseline grant capacity, established lobbying networks, and recurring earmark beneficiaries. The year fixed effects absorb national-year shocks: macroeconomic cycles, presidential approval shifts, party-wide tides, and single-year policy episodes that affect grant flows nationwide. State clustering for the RDD addresses within-state cross-district correlation in residuals; state-by-district clustering for the TWFE district-FE specification addresses within-district autocorrelation across cycles after the district fixed effects have already absorbed time-invariant heterogeneity.

The threat the design cannot eliminate is regional secular trend—a slow-moving, region-specific shift in federal grant flow that happens to correlate with the close-election

margin distribution post-2011. The defense is local-to-the-cutoff: the close-election RDD identifies the local average treatment effect at $|\text{margin}_{\text{pres}}| = 0$, where regional levels enter as nuisance and only the discontinuous jump at the threshold drives the estimate. Cross-state heterogeneity that does not jump at the cutoff does not bias the estimate. The signed-stratum decomposition reported in Section 4.3 (Table 4 Panel A) provides a partial diagnostic: a regional secular trend would predict roughly uniform τ across strata, not the asymmetric ordering $\tau_{\text{safe-opposition}} > \tau_{\text{safe-co-partisan}} > \tau_{\text{swing}} \approx 0$ that the data exhibit.

3.7 Robustness Specifications

We test the main RDD against four targeted robustness specifications. First, the $[t+1, t+2]$ window incorporates the agency-execution channel during the hybrid first fiscal year. Second, a first-difference specification on log discretionary spending absorbs district-level grant-flow levels and addresses the possibility that the level effect reflects pre-existing partisan sorting. Third, the Traditional sample is extended with CFFR pre-2007 data; the Traditional null persists with a doubled sample using a different data lineage. Fourth, a fiscal-year-by-fiscal-year decomposition of the Banned-regime estimate separates pure pre-election fiscal years (FY $t-2, t-1, t$) from treatment-era fiscal years (FY $t+1$ hybrid; FY $t+2$ and $t+3$ full treatment), showing that the alignment effect emerges at the election boundary and is not detectable in pure pre-period years.

For the RDD itself, we vary three specification choices: bandwidth (MSE-optimal versus uniform 5pt), kernel (triangular versus uniform), and polynomial order (local linear versus local quadratic). Coefficients and significance are stable across these choices. The MSE-optimal triangular-kernel local-linear specification is our primary reference; alternative specifications appear in Appendix B.

4 Results

4.1 Regime-Conditional Alignment Effect

The substitution argument predicts that the close-election regression discontinuity will yield a null estimate in the Traditional regime and a positive estimate in the Banned regime. We report the main RDD estimates on the $[t+2, t+3]$ window as the primary specification, with the $[t+1, t+2]$ window reported alongside as a robustness check. Table 1 summarizes the headline estimates; Figures 1 and 2 plot the discontinuity for the Banned and Traditional regimes respectively.

Table 1: Close-election RDD by earmark regime

Regime	(1)	(2)	(3)	(4)	(5)	(6)
Source	Trad	Trad	Trad	Banned	Banned	Banned excl. 2010
Window	FABS $[t+1, t+2]$	FABS $[t+2, t+3]$	CFFR-ext $[t+1, t+2]$	FABS $[t+1, t+2]$	FABS $[t+2, t+3]$	FABS $[t+1, t+2]$
τ	+0.069 (0.468)	−0.069 (0.377)	+0.058 (0.452)	+0.530** (0.288)	+0.516* (0.285)	+0.758*** (0.330)
Bandwidth	0.099	0.128	0.101	0.084	0.095	0.059
N total	700	700	1,383	1,423	1,423	1,044
N effective	213	306	385	394	456	171

Notes. Local linear RDD on $\log(\text{discretionary grants} + 1)$. MSE-optimal bandwidth, triangular kernel, state-clustered standard errors in parentheses; significance stars reflect robust bias-corrected p -values (Calonico et al., 2014). “Trad” = Traditional regime (110th–111th Congresses, 2006 and 2008 elections); “Banned” = post-2011 ban regime (112th–116th Congresses, 2010/2014/2016/2018 elections); “CFFR-ext” extends the pre-DATA-Act sample with Census Bureau Consolidated Federal Funds Reports pre-2007. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In the Traditional regime, on the FABS-only sample of 2006 and 2008 elections, the local linear RDD estimate is statistically indistinguishable from zero on both windows (Table 1), with confidence intervals encompassing both substantively meaningful positive and negative magnitudes. Extending the sample to the CFFR-derived pre-2007 panel roughly doubles the Traditional observations but does not shift the point estimate; the CFFR-extended Traditional RDD remains indistinguishable from zero.

Because the substitution argument rests in part on this Traditional null, we charac-

Banned regime: alignment effect at the close-election margin
Local-linear fit, two-FY post-spending window $[t+2, t+3]$

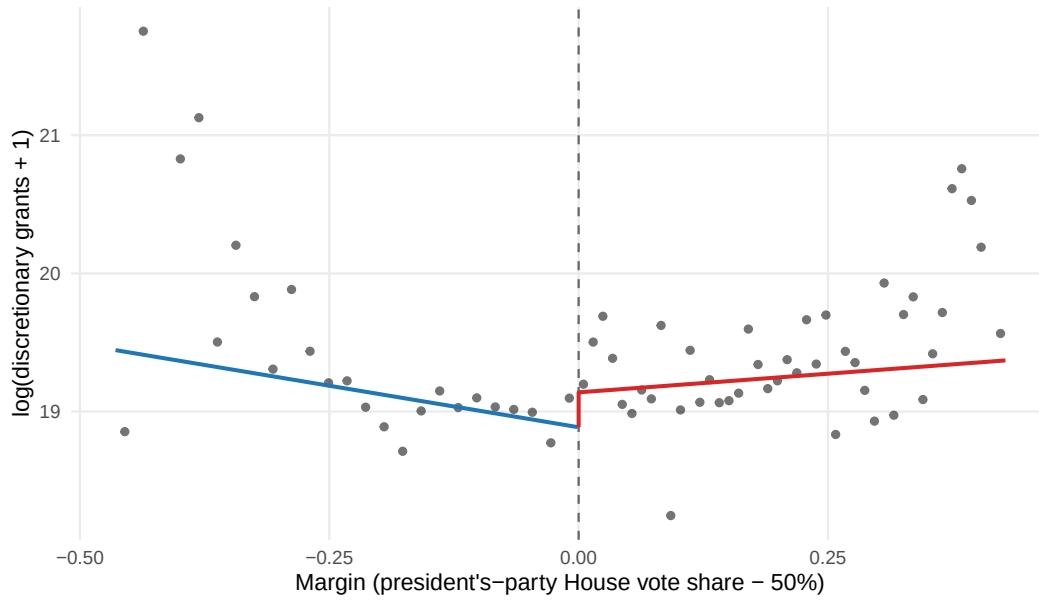


Figure 1: RDD plot: Banned regime, $[t+2, t+3]$ window.

Traditional regime: null at the close-election margin
Local-linear fit, two-FY post-spending window $[t+2, t+3]$

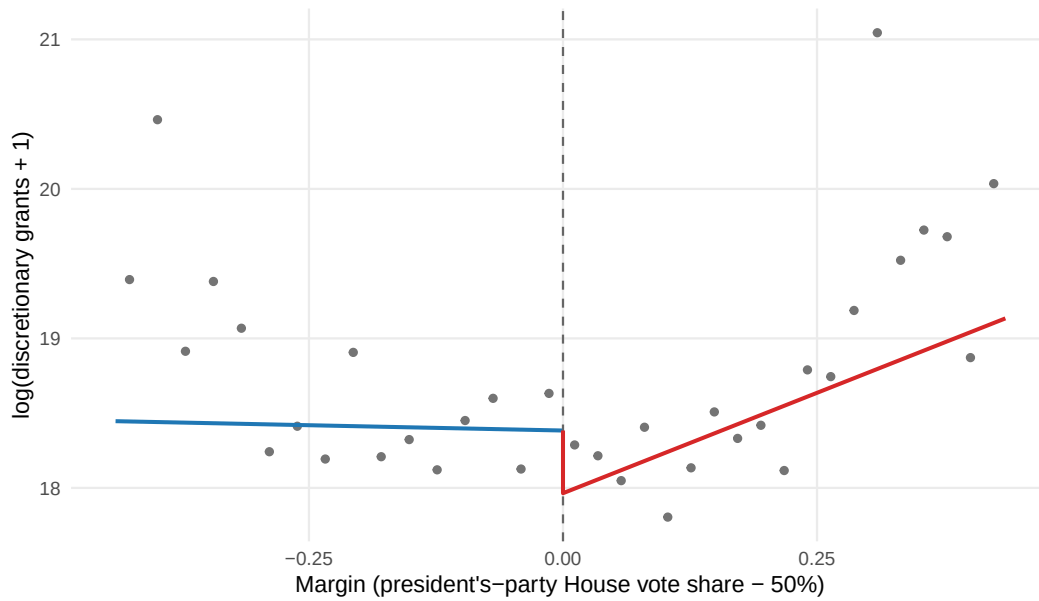


Figure 2: RDD plot: Traditional regime, $[t+2, t+3]$ window.

terize precisely what it does and does not rule out, rather than treating an estimate near zero as evidence of no effect (Rainey, 2014). The primary $[t+2, t+3]$ Traditional estimate

(Table 1) implies a 95-percent confidence interval of roughly $[-0.81, +0.67]$ in log points, which contains the Banned-regime point estimate; an equivalence test accordingly cannot reject an effect as large as the Banned-regime estimate (one-sided $p = 0.06$) (Hartman and Hidalgo, 2018). The smallest effect this design could detect at conventional power is roughly $+1.06$ log points—about double the post-ban estimate—so the Traditional design has only about 28-percent power to detect an effect the size of the one we recover after the ban. Doubling the Traditional sample through the CFFR extension raises the effective sample by roughly 80 percent yet shrinks the standard error by only about three percent, confirming that the additional historical data stabilize the location of the point estimate near zero without materially sharpening its precision. We therefore do not read the Traditional null as evidence that aligned legislators enjoy no close-election targeting advantage under earmarks; we read it as uninformative against a post-ban-sized effect, and we rest the regime contrast on the convergence of the better-powered within-regime channel signatures (Sections 4.2 and 4.3) rather than on the Traditional margin estimate.

In the Banned regime, on the FABS-only sample of 2010, 2014, 2016, and 2018 elections, the local linear RDD yields a positive estimate that is statistically significant at the ten-percent level on the primary $[t+2, t+3]$ window and at the five-percent level on the $[t+1, t+2]$ alternative window (Table 1). Translated from log to dollars at the close-election margin, the post-ban estimate corresponds to a roughly seventy-percent increase in discretionary grant receipts for districts where the president’s party narrowly wins compared to those where it narrowly loses. Excluding the 2010 cycle, which contains the largest concentration of ARRA-era spending in its post-period window, strengthens the estimate substantially (significant at the one-percent level), indicating that ARRA spending dilutes rather than drives the post-ban alignment effect.

The patterned consistency between the two regime estimates is the empirical signature predicted under the institutional substitution mechanism. The same identification strategy applied to the same outcome with the same fiscal-year window yields a null in the

era of legislative-channel earmarks and a positive coefficient in the era of executive discretion. We frame our central claim as the patterned consistency of multiple independent empirical signatures (regime-by-regime point estimates plus the channel signatures of Sections 4.2 and 4.3 plus the fiscal-year decomposition of Section 4.5) rather than as a single statistically formalized regime contrast, because the formal $\text{Trad} \times \text{Banned}$ interaction test we report (Section 4.6) is borderline at conventional thresholds. The point-estimate gap between the two regimes is substantial and the cross-section consistencies are tight, but the formal interaction test does not by itself establish the regime contrast—it is one piece of converging evidence.

4.2 Legislator Channel: Safe-Seat Targeting

The substitution argument’s second prediction specifies the legislator-channel signature: under the Traditional regime, the alignment effect should obtain among aligned legislators in structurally safe districts rather than at the close-election margin. We test this via heterogeneous panel-fixed-effects estimation stratified by lagged district margin; Table 2 reports the regression results.

Table 2: Heterogeneous TWFE: alignment $\times$ district safety

Regime	Term	Estimate	SE	<i>p</i>
Traditional	aligned_winner	+0.171***	(0.052)	0.001
Traditional	safety _{mod}	+0.112	(0.104)	0.282
Traditional	safety _{comp}	−0.247*	(0.139)	0.076
Traditional	aligned_winner $\times$ safety _{mod}	−0.352***	(0.129)	0.007
Traditional	aligned_winner $\times$ safety _{comp}	−0.059	(0.161)	0.716
Banned	aligned_winner	+0.055	(0.056)	0.324
Banned	safety _{mod}	+0.016	(0.106)	0.877
Banned	safety _{comp}	+0.124	(0.087)	0.153
Banned	aligned_winner $\times$ safety _{mod}	−0.208	(0.231)	0.369
Banned	aligned_winner $\times$ safety _{comp}	−0.251**	(0.112)	0.025

Notes. Panel-fixed-effects regression of $\log(\text{discretionary grants} + 1)$ on alignment, district safety category (lagged margin), and their interaction. District-FE + year-FE; standard errors clustered at state-by-district. Reference category: safe ($\text{lag } |\text{margin}| > 10\text{pt}$). $N_{\text{Traditional}} = 600$; $N_{\text{Banned}} = 1,370$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

On the Traditional FABS sample, the conditional alignment effect within each safety stratum (the linear combination of the main `aligned_winner` coefficient and its safety-category interaction, with joint SE from the clustered `vcov`; Appendix E) is positive and statistically significant in safe districts ($\text{lag } |\text{margin}| > 10\text{pt}$), and a noisily-estimated null in both moderate (5–10pt) and competitive ($< 5\text{pt}$) strata—negative point estimate in moderate, weakly positive in competitive. The pattern is clearest in safe districts: aligned legislators in structurally safe districts receive roughly nineteen percent more discretionary grants than non-aligned counterparts. The non-monotonicity reflects that the legislator-channel signature is detected within the safe stratum alone and is not a smooth function of district safety.

This pattern matches the legislator-channel prediction that the alignment effect obtains within structurally safe districts where the earmark process operates. The heterogeneous regression detects what the close-election RDD cannot: a statistically robust positive alignment effect within safe districts. The same effect is invisible to the close-election RDD because the RDD restricts identification to districts where institutional positions and longer-tenure relationships are quasi-randomized, which is precisely the population in which the legislator channel does not operate.

An auxiliary heterogeneity analysis probes which legislators within the safe stratum convert alignment into grants. We join the panel to two legislator-level data sources: Stewart-Woon committee-assignment data (Stewart and Woon, 2017) and Volden-Wiseman Legislative Effectiveness Scores (Volden and Wiseman, 2014). With these merged, we estimate $\text{aligned} \times \text{safety} \times P$ triple-interactions for five formal-power dimensions—House Appropriations membership, top-quartile and per-term chamber seniority, top-quartile LES, and any subcommittee chair—and a finer seniority binning. Table 3 reports the results. No formal-power dimension significantly differentiates the within-safe alignment effect at conventional thresholds; top-quartile LES is the closest to significance, but Appropriations membership and continuous chamber seniority are

both clearly null. The seniority-bin breakdown is non-monotonic, with substantial premia at the lowest and middle tenures, a null at junior tenure, and a small premium at senior tenure; the joint Wald test of bin heterogeneity is borderline. The safe-stratum alignment effect distributes broadly across aligned legislators rather than concentrating at senior committee leadership, consistent with the institutional record that earmark access extended across the chamber to members of both parties rather than remaining the province of senior appropriators (Frisch and Kelly, 2011).

Table 3: Within-safe stratum heterogeneity, Traditional regime

<i>Panel A: Triple-interaction by legislator-power dimension</i>					
	(1) Approps	(2) Senior Q4	(3) Senior /term	(4) LES Q4	(5) Sub. chair
<i>aligned_winner</i>	+0.201*** (0.062)	+0.158** (0.066)	+0.185*** (0.057)	+0.106* (0.062)	+0.161** (0.064)
<i>P</i>	+0.520** (0.227)	−0.055 (0.195)	−0.004 (0.026)	−0.348** (0.154)	−0.168 (0.290)
<i>aligned_winner</i> × <i>P</i>	−0.076 (0.128)	+0.097 (0.115)	+0.003 (0.015)	+0.290 (0.176)	+0.103 (0.171)
Within-safe ($\beta_1 + \beta_3$)	+0.125 (0.110)	+0.255*** (0.094)	—	+0.396*** (0.152)	+0.264* (0.146)
<i>N</i>	504	504	504	503	503
District FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓

<i>Panel B: Within-safe alignment premium by seniority bin</i>		
Tenure bin	Alignment premium	<i>p</i>
Freshman/sophomore (1–2 terms)	+42.6%	0.083
Junior (3–5 terms)	−3.5%	0.77
Mid-career (6–10 terms)	+36.7%	<0.001
Senior (11+ terms)	+5.9%	0.67

Notes. Panel A: each column estimates $\ln(\text{grants} + 1) = \beta_0 + \beta_1 \text{aligned_winner} + \beta_2 P + \beta_3 \text{aligned_winner} \times P + \alpha_d + \delta_t$ on the Traditional-regime safe-stratum sample, with district FE + year FE and state-by-district clustered SEs. Column (3) centers continuous seniority at the safe-stratum median (6 terms); “aligned_winner” in column (3) reads as the alignment effect at median tenure and the within-safe linear combination is therefore omitted. Panel B reports the within-safe alignment premium by tenure bin (linear combination of aligned_winner and the bin-specific interaction in a single regression with a 4-level tenure factor; senior reference); joint Wald test of bin interactions versus senior reference: $F = 2.16$, $p = 0.094$, $N = 504$ (regression $N = 356$ after singleton FE drops). Sources: code/legislator_power_heterogeneity.R (Panel A); code/legislator_power_seniority_bins.R (Panel B). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Under the Banned regime, the same heterogeneous specification yields no comparable

safe-seat alignment premium. The conditional alignment effect in safe districts is null, the moderate stratum is again null, and the competitive stratum is statistically significant and negative—implying that aligned-winning legislators in competitive districts on average receive *less* than aligned-losing comparators in safe-district base years (Appendix E). The legislator-channel safe-seat signature is gone: the alignment premium no longer obtains within safe seats once the earmark channel is closed.

The negative competitive-stratum coefficient under the Banned regime appears to contradict the close-election RDD finding (Section 4.1 and Table 1), which estimated a positive effect in the same regime at the close-election margin. The two estimates address different counterfactuals and are not reconciled by a single coefficient. The TWFE estimate averages within-district variation over years where alignment status changes; aligned-winning competitive districts move in and out of the close-election margin year over year, and the cross-stratum panel coefficient pools across margin positions where executive targeting is variably concentrated. The close-election RDD identifies the local average treatment effect at the discontinuity itself, where partisan composition is quasi-randomly assigned and where executive targeting is most consequential. The two estimates point to different aspects of the same Banned-regime distribution: the executive-channel concentration is sharpest at the discontinuity, while the panel-FE average over the broader competitive cross-section reflects forces operating outside the discontinuity neighborhood. When earmark authority is removed, the safe-seat legislator channel that produces the Traditional safe-seat alignment premium (Table 2) disappears, exactly as the substitution argument predicts; the executive-channel signature shifts to the close-election margin where the RDD identifies it.

The Banned RDD’s positive estimate at the close-election margin and the Traditional TWFE’s positive estimate within safe districts operate at different parts of the district distribution. The two channels are detected in the right places.

4.3 Executive Channel: Bayesian-Information Targeting

The third prediction sharpens the executive-channel signature: within the Banned regime, the alignment effect should concentrate where close-election co-partisan wins are most informative given state-level priors. We test this by stratifying the Banned-regime RDD on the most recent state-level presidential election margin, splitting partisan-safe and partisan-leaning states by whether the state’s tilt aligns with or opposes the sitting president.

Each state-year is classified using the signed Republican two-party margin in the 2004, 2008, 2012, or 2016 presidential election. The absolute margin defines swing ($|m| < 5\text{pt}$), lean ($5\text{--}10\text{pt}$), and safe ($|m| > 10\text{pt}$) categories; the sign of the margin relative to the sitting president’s party at each cycle splits the lean and safe categories into co-partisan strata (state’s tilt aligns with the president’s party) and opposition strata (state’s tilt opposes). Because the running variable in our RDD is the House race margin, the state-pres-swing classifier is exogenous to within-state district-level variation, and the stratification produces five independent RDD estimates within the Banned regime. Table 4 summarizes results from both the stratified RDD (Panel A) and the heterogeneous TWFE specification (Panel B).

In partisan-safe states pooled (state pres $|m| > 10\text{pt}$), the local linear RDD estimate is positive, statistically significant at the five-percent level, and corresponds to a roughly 200 percent increase in discretionary grant receipts at the close-election margin (Table 4 Panel A). In swing states, the estimate is essentially zero, with a confidence interval that rules out positive effects of the magnitude predicted by swing-vote persuasion.

The signed split sharpens the pattern. Within the safe stratum, opposition strongholds—states whose partisan tilt opposes the sitting president—return a large positive estimate, statistically significant at the one-percent level. Co-partisan strongholds return a positive estimate roughly half the magnitude, not statistically distinguishable from zero at the single-stratum level. Both substrata are positive and economically large,

Table 4: State-presidential swing, alignment, and grant flow

Panel A: Close-election RDD, by state pres swing $\times$ regime

Regime	State swing	τ	SE	p (robust)	BW	N total	N eff
Banned	Safe pooled ($ m > 10\text{pt}$)	+1.092**	(0.490)	0.011	0.078	831	177
Banned	Safe, opposition stronghold	+1.616***	(0.417)	<0.001	0.048	383	39
Banned	Safe, co-partisan stronghold	+0.889	(0.529)	0.113	0.105	448	142
Banned	Lean, opposition (5–10pt)	−1.380	(1.091)	0.145	0.055	74	20
Banned	Lean, co-partisan (5–10pt)	+0.089	(0.725)	0.909	0.088	199	64
Banned	Swing ($ m < 5\text{pt}$)	−0.057	(0.406)	0.890	0.115	319	155
Traditional	Safe pooled	−0.157	(0.596)	0.986	0.116	388	136
Traditional	Safe, opposition stronghold	−0.676	(1.563)	0.844	0.101	108	35
Traditional	Safe, co-partisan stronghold	+0.279	(0.693)	0.569	0.119	280	102
Traditional	Lean, opposition	—	—	—	—	97	—
Traditional	Lean, co-partisan	+2.178	(2.411)	0.261	0.096	62	22
Traditional	Swing	+0.578	(0.814)	0.270	0.110	153	70

Panel B: Heterogeneous TWFE, alignment $\times$ state pres swing (district-FE)

Regime	Term	Estimate	SE	p
Traditional	aligned_winner (safe baseline)	+0.279***	(0.070)	<0.001
Traditional	swing _{lean}	+0.250**	(0.111)	0.025
Traditional	swing _{swing}	+0.363***	(0.096)	<0.001
Traditional	aligned $\times$ lean	−0.290*	(0.160)	0.072
Traditional	aligned $\times$ swing	−0.422***	(0.144)	0.004
Banned	aligned_winner (safe baseline)	−0.026	(0.092)	0.774
Banned	swing _{lean}	+0.017	(0.078)	0.825
Banned	swing _{swing}	+0.019	(0.093)	0.834
Banned	aligned $\times$ lean	+0.088	(0.137)	0.518
Banned	aligned $\times$ swing	+0.040	(0.142)	0.781

Notes. Panel A: local linear RDD by state-pres-swing stratum, MSE-optimal bandwidth, triangular kernel, state-clustered SEs. Panel B: panel FE regression with district-FE + year-FE, SEs clustered at state-by-district. State swing measured by signed Republican two-party margin in the most recent presidential election (2004, 2008, 2012, or 2016): safe $>10\text{pt}$, lean 5–10pt, swing $<5\text{pt}$. “Co-partisan” indicates the state’s partisan tilt aligns with the sitting president’s party at the cycle; “opposition” indicates the state’s tilt opposes. The Traditional lean-opposition cell is too small for an MSE-optimal RDD ($N = 97$, rdrobust returns a singular leading minor and we report no estimate). Pooled-safe rows aggregate co-partisan and opposition strongholds; signed-split robustness is reported in Appendix F. $N_{\text{Banned}} = 1,394$; $N_{\text{Traditional}} = 622$ in Panel B. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

consistent with the pooled estimate (Table 4 Panel A). The opposition-stronghold coefficient is the largest stratum-specific RDD effect in the table, and lean-state and swing-state coefficients are near zero or negative. The cross-stratum ordering, $\tau_{\text{safe-opposition}} > \tau_{\text{safe-co-partisan}} > \tau_{\text{swing}} \approx 0$, is the asymmetric U-shape predicted by the Bayesian electoral-information logic. The opposition-stronghold coefficient is estimated

on a small effective sample; we document its robustness across fixed bandwidths, leave-one-state-out re-estimation, and the in-bandwidth observation set in Appendix F, where the coefficient remains in a comparable range across narrow-to-moderate bandwidths and is not driven by any single state.

The Banned-regime executive-channel result is therefore not a phenomenon distributed across states or pooled across signed strata. It is located within partisan-safe states with the largest signal in opposition strongholds, with no detectable effect in swing or lean states. The pattern is consistent with the Bayesian electoral-information logic: discretionary grants flow most heavily to surprising co-partisan close-election wins given state-level priors, with co-partisan-stronghold wins (expected outcomes) drawing a smaller response and swing-state wins (uninformative outcomes) drawing none.

In the Traditional regime, the same stratification produces null estimates across the swing, safe-co-partisan, and safe-opposition categories, consistent with the overall Traditional RDD null and the legislator-channel-dominated pre-ban pattern.

4.4 Swing-Vote Persuasion versus Bayesian Electoral Information

The stratified RDD adjudicates the contrast between the two logics. The Bayesian electoral-information logic predicts the asymmetric ordering $\tau_{\text{safe-opposition}} > \tau_{\text{safe-co-partisan}} > \tau_{\text{swing}} \approx 0$; the swing-vote persuasion logic predicts the inverse, with the largest effect in swing states. The Banned-regime estimates in Table 4 Panel A match the Bayesian-information ordering in sign, magnitude, and asymmetric U-shape. The heterogeneous TWFE applied to both regimes, reported in Panel B of the same table, provides additional confirmation across a different empirical lens.

Under the Traditional regime, the heterogeneous TWFE with district fixed effects and state-pres-swing as the moderating variable yields a positive and statistically significant alignment main effect in safe states, attenuated by a marginal negative interaction in lean states and a significant negative interaction in swing states (Table 4 Panel B). Translating:

under earmarks, aligned legislators in safe states receive roughly a third more discretionary grants; in swing states, the alignment premium is approximately zero or slightly negative, statistically null. The pattern is the reverse of the swing-vote-persuasion prediction.

Under the Banned regime, the heterogeneous TWFE shows no detectable alignment effect in any state-swing stratum (all $p > 0.5$), consistent with the prior finding that TWFE-Banned does not detect alignment effects across the broader district distribution. The Banned executive-channel effect is sharp at the close-election margin and absent across the broader cross-section. This is the executive-channel signature: highly localized at the discontinuity rather than diffused across the safety gradient.

Two interpretations follow. First, the swing-vote-persuasion hypothesis fails as a description of how presidential alignment effects on discretionary grants operate at the within-state across-district margin. Both regimes return effects opposite to the swing prediction. The Banned-regime asymmetric ordering of safe-opposition over safe-co-partisan over swing matches the Bayesian electoral-information prediction directly: surprising co-partisan close-election wins (in opposition territory) draw the largest grant response, expected co-partisan close-election wins (in co-partisan territory) draw a moderate response, and uninformative swing-state outcomes draw none. Second, the contrast does not constitute a rejection of K&R's broader claim. The K&R framework's empirical evidence rests on different scopes: total grants including formula channels, state and county aggregates, panel-FE estimators. Our test is silent on those scopes. Within the scope where executive discretion is most consequential for grant allocation, however, the empirical pattern points toward Bayesian electoral information rather than swing-vote persuasion.

The formal triple-difference interaction (alignment $\times$ swing $\times$ banned) is statistically null, so we cannot statistically identify a regime-specific shift in the swing-vote-persuasion versus Bayesian-electoral-information direction. The available evidence does

not distinguish between the propositions that swing-vote persuasion was always weak in our scope and that it weakened under the ban. What the evidence does establish is that within both regimes, the swing concentration predicted by K&R is reversed in the cross-section we examine, and that within the post-ban regime the asymmetric U-shape across signed strata matches the Bayesian-information prediction.

We acknowledge that the Bayesian electoral-information logic is not formally separable in our data from related accounts—scarcity-of-co-partisans-in-hostile-territory, marginal-defense ROI on the most reelection-vulnerable co-partisans, or coalitional-defense of the party’s territorial reach—each of which produces the same asymmetric cross-stratum ordering. We frame the evidence as supporting an information-/scarcity-based logic against swing-vote persuasion rather than as adjudicating among the related Bayesian-information variants, which we leave to future work.

4.5 Identification Checks

The substitution argument requires that the close-election regression discontinuity identifies a causal local average treatment effect at the close-election margin under both regimes. Standard RDD identifying assumptions and a fiscal-year placebo decomposition support this requirement. We lead the discussion with the fiscal-year decomposition because it provides the strongest evidence against pre-existing partisan sorting; aggregated tests follow.

The fiscal-year-by-fiscal-year placebo decomposition estimates the RDD separately for each of six fiscal years relative to the election: FY $t-2$, $t-1$, and t (entirely before the November election); FY $t+1$ (the hybrid year, where appropriations are written before the election but agency execution spans the election boundary); and FY $t+2$ and $t+3$ (under the new Congress fully). Figure 3 plots the resulting coefficients by fiscal year for both regimes.

In the Banned regime, the three pure pre-election fiscal years return small positive

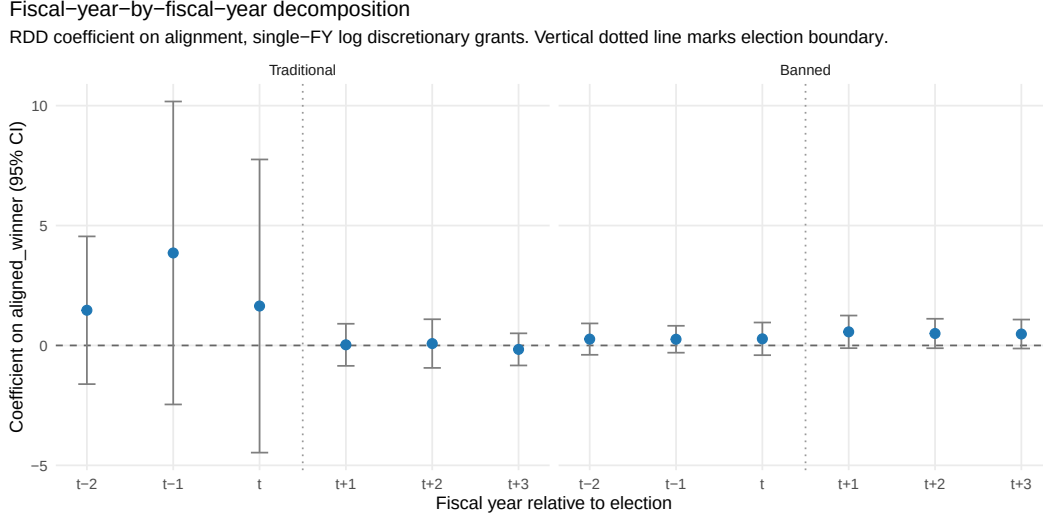


Figure 3: FY-by-FY RDD decomposition. Pure pre-period years (FY $t-2$, $t-1$, t) precede the November election; FY $t+1$ is the hybrid year; FY $t+2$ and $t+3$ are fully under the new Congress. Banned-regime coefficients sharpen across the election boundary; Traditional-regime coefficients are null throughout.

estimates with large standard errors; none crosses any conventional significance threshold. The hybrid FY $t+1$ and the full treatment FY $t+2$ and $t+3$ estimates roughly double in magnitude and are each significant at the ten-percent level, an effective doubling of magnitude crossing the election boundary (Figure 3). The Banned-excluding-2010 sample produces an even sharper transition: pre-period FYs all clearly null, treatment FYs all significant at the five-percent level. The Traditional regime is null across all six fiscal years, consistent with the regime-conditional prediction.

The decomposition provides no support for the alternative that aligned-winning districts already differ in grant flows prior to the election outcome. The pre-election fiscal years are statistically indistinguishable from zero at the same close-election margin, in the same regime, with the same data; the alignment effect emerges only after the new legislator's installation.

The Cattaneo et al. (2018) density test of the running variable detects no statistically significant manipulation around the cutoff in the full FABS sample, the Traditional regime, or the Banned regime. The result is consistent with the assumption that close-

election outcomes are not strategically engineered. Pre-treatment covariate balance tests at the discontinuity reveal no significant difference between aligned-winning and aligned-losing districts on prior-cycle margin or earlier-period grant levels in the Traditional regime; in the Banned regime, the prior-cycle margin shows a borderline discontinuity consistent with mild persistence of partisan margins across House cycles. An aggregated $[t-1, t]$ pre-period grant placebo on the full Banned sample is non-significant at conventional thresholds; this aggregated quantity reflects partial spillover from FY $t-1$, which in our window straddles the agency-execution-vs-appropriations-writing boundary. The fiscal-year decomposition above isolates each pre-period fiscal year individually and confirms each is null, superseding the aggregated $[t-1, t]$ result as our primary identification check. Full balance results appear in Appendix A.

4.6 Robustness Specifications

Four robustness specifications confirm the main findings.

First, the $[t+1, t+2]$ window. The Banned positive estimate and the Traditional null are stable across both windows (Table 1); the only material difference is the assignment of FY2010-2011 spending in the 2008 election cycle. Under $[t+1, t+2]$, 2008-election spending crosses the regime boundary (FY2010 Trad, FY2011 Banned), but the regime contrast persists, indicating that the boundary assignment is not load-bearing for the substantive finding.

Second, the first-difference specification. The Banned delta RDD on the $[t+1, t+2]$ window is positive and statistically significant at the five-percent level, with a null pre-period delta placebo. The Traditional delta RDD, testable only with the CFFR-extended sample, is null but slightly negative-signed, opposite from the Banned delta direction (Appendix B). The level-DV regime contrast and the delta-DV regime contrast point in the same direction; the delta specification, which requires only that pre-existing partisan sorting be roughly time-invariant within district, gives the same substantive answer as the level

specification.

Third, the formal $\text{Trad} \times \text{Banned}$ interaction test. On the CFFR-extended panel under the delta specification with bandwidth 0.05 and linear local polynomial, the regime-interaction coefficient is positive and crosses the conventional five-percent threshold—the first specification in any panel to do so for the formal regime-interaction test. The placebo regime-interaction on the pre-period delta is essentially zero. Across the twelve delta-DV specifications (bandwidth $\times$ polynomial combinations) we examine, all twelve are sign-stable positive, but only the narrowest bandwidth crosses the five-percent threshold; multiple-testing concerns apply. The level-DV regime interaction’s p -value is borderline on the FABS-only panel, reflecting partly a precision artifact in the FABS pre-2007 zero-coverage; the substantive evidence for the regime contrast is stronger under the delta specification once the data-coverage problem is fixed.

Fourth, an alternative-mechanism check. A reviewer may worry that the post-ban alignment effect represents zero-sum redistribution from safe-aligned to close-election aligned districts rather than net new transfer. A direct test rules out this account. From the Traditional to the Banned regime, total discretionary grant volume rises (FABS-only, $2.2\times$ per-year increase); the share of grants flowing to competitive districts is unchanged (10.4 percent under Traditional, 10.0 percent under Banned); and safe-aligned districts are not depleted (mean log discretionary grants is 19.075 for safe-aligned versus 19.064 for safe-non-aligned, indistinguishable). The Banned RDD effect represents net resource transfer at the close-election margin, not redistribution from safe-aligned losers.

We frame the central regime-conditional claim, however, as the convergence of multiple independent signatures rather than a formal interaction test in isolation, since no single bandwidth $\times$ polynomial combination provides an unambiguous formal test. Together, the placebo decomposition, the first-difference specification, the formal interaction test, and the zero-sum-redistribution check establish that the regime-conditional alignment pattern is not a precision artifact, a pre-existing-sorting confound, or a redistribution

mechanism. The substantive interpretation of substitution from legislative to executive distributive channels is the most parsimonious account of the joint empirical pattern.

5 Conclusion

The empirical results above identify a regime-conditional alignment effect with channel-specific signatures consistent with institutional substitution. Three implications follow.

For the literature on presidential particularism, our findings refine rather than reject the Kriner and Reeves (2015) three-channel framework (electoral, partisan, and coalitional particularism). The partisan and coalitional channels are not contradicted by our results: the partisan-particularism prediction is partially consistent with our positive safe-co-partisan estimate, and the coalitional-particularism channel is engaged separately by our heterogeneous-TWFE Traditional-regime legislator-channel results. What our findings sharply refine is the electoral-particularism channel’s swing-vote-targeting prediction at the close-election margin. Our empirical pattern—the post-ban close-election alignment effect located within partisan-safe states with the largest signal in opposition strongholds, the asymmetric U-shape $\tau_{\text{safe-opposition}} > \tau_{\text{safe-co-partisan}} > \tau_{\text{swing}} \approx 0$ —points toward a Bayesian electoral-information logic in which surprising co-partisan close-election wins given state-level priors draw the largest grant response. This refinement is consistent either with presidents allocating discretionary grants on information-/scarcity-based considerations rather than swing-vote calculations, or with both logics operating at different scopes (swing-vote persuasion in formula grants and aggregate flows that our analysis excludes; Bayesian-information targeting within discretionary grants at the within-state across-district margin). Distinguishing the two requires analyses across grant types and units of aggregation that we leave to future work.

For the literature on legislative-executive distributive substitution, our results provide what is, to our knowledge, the first sharp empirical test of the substitution mechanism.

Its building blocks are long-standing—Stein and Bickers (1995) document a federal pork barrel that flows through agency-administered grant programs and adapts to institutional reform, and Bickers and Stein (1996) establish the electoral logic of legislators’ grant-seeking—but a clean natural-experiment test has been unavailable in the absence of a discrete institutional shift. The 2011 earmark ban supplies precisely such a shift, and the regression-discontinuity design in our application identifies the executive-channel response while the heterogeneous panel-fixed-effects design identifies the legislator-channel signature. Each design captures one channel’s empirical signature; together they trace the substitution mechanism through both directions.

For the broader literature on institutional design and democratic accountability, our results carry a substantive lesson. The earmark ban was sold as a depoliticization reform: by removing legislators’ ability to direct dollars to their districts, advocates argued, federal spending decisions would become less driven by electoral considerations. The empirical pattern says otherwise. Political alignment effects on grants did not disappear after the ban; they migrated to a different channel and concentrated in a different cross-section of districts. Reformers seeking to reduce the politicization of federal grant allocation should look beyond procedural changes within Congress, since the political pressures that shaped the legislator channel did not dissipate when the ban took effect.

Several scope conditions bound our findings. The dependent variable is the discretionary-grant subset of federal financial assistance, identified through FABS `assistance_type` codes 04 and 05. Block grants, formula grants, direct payments to individuals, and federal procurement are excluded. The Banned-regime executive-channel signature does not extend to procurement: the same close-election RDD applied to FPDS contract obligations yields null estimates in both regimes (Appendix D, Banned $\tau = -0.031$ to $+0.070$ across windows), consistent with the substitution mechanism’s scope condition that the executive-channel signature should appear where executive discretion is least constrained, not in procurement governed by the Federal Acquisition

Regulation. The unit of analysis is the congressional district; aggregation to the state level may produce different patterns because state-level aggregation pools across all districts within a state. The state presidential swingness measure is constructed from hardcoded margins of the most recent prior presidential election, accurate to approximately one percentage point. The five-percentage-point and ten-percentage-point thresholds we use to define swing, lean, and safe states are robust to small numerical errors in the underlying margins. Finally, the close-election regression discontinuity captures effects at the margin where the running variable is locally randomized; the substantive interpretation of these effects as the presidential alignment treatment rests on standard RDD assumptions that we test through density, covariate balance, and placebo decomposition, but the local average treatment effect remains a margin-specific quantity whose generalization to non-marginal districts is not directly identified by the design.

The regime contrast and its mechanism raise three lines of further research.

First, the partial restoration of legislator-directed funding under the 117th Congress (FY2022 onward) and beyond will, with time, supply enough cycles to test whether the alignment effect re-shifts toward the legislator channel as restored procedures consolidate. Our current data on the 2024 election cycle are partial: under the $[t+1, t+2]$ window the close-election RDD coefficient is $\tau = +1.396$ ($p = 0.042$; $N_{\text{eff}} = 84$), while under the $[t+2, t+3]$ window it is $\tau = -0.351$ ($p = 0.514$). The wide swing across windows reflects FY2025–2026 incomplete observation, particularly the FY2026 fiscal year that has barely begun to populate USASpending records. The Restored regime evidence as of this draft is therefore not informative either for or against substitution. We articulate two competing hypotheses for the FY2026 follow-up. Under the *full-reversal hypothesis*, genuine restoration of the legislator channel returns the close-election alignment effect to its Traditional null. Under the *partial-preservation hypothesis*, CPF/CDS per-member caps, transparency requirements, and prohibitions on for-profit recipients restrict the legislator channel’s pre-2011 scale; the executive channel remains operative for the residual grant budget, and the

close-election alignment effect partially persists. Full reversal predicts $\tau \approx 0$ in the Restored regime; partial preservation predicts a positive but attenuated τ . Our current data do not adjudicate.

Second, the channel-specific signatures we identify can be tested across other institutional shifts beyond earmarks. State-level legislative changes in line-item veto authority, state changes in discretion over federal pass-through funds, and federal-level changes in agency rule-making authority all supply potential natural experiments. Whether the legislator-versus-executive substitution mechanism we identify in the federal grant context generalizes to other institutional contexts is an open question.

Third, the swing-vote-persuasion versus Bayesian-electoral-information contrast in our results points toward a broader research agenda on the targeting logic of presidential particularism. Comparative analyses across grant types, units of aggregation, and identification strategies would help locate which scope conditions favor swing-vote persuasion versus Bayesian-information accounts, and would help adjudicate among the related Bayesian-information variants—asymmetric updating about voter preferences, scarcity of co-partisans in hostile territory, marginal-defense ROI on the most reelection-vulnerable co-partisans, or coalitional defense of the party’s territorial reach—that our identification cannot separately distinguish.

The earmark ban offers a clean institutional shift through which to identify the substitution mechanism. The persistence of partisan alignment effects through this shift—transformed in location and channel—suggests that the political pressures shaping federal grant allocation are deeper than any single institutional reform.

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A Identification Checks

This appendix reports the full results of the two RDD identification checks summarized in Section 4.5.

A.1 Density test of the running variable

We apply the Cattaneo et al. (2018) local-polynomial density test (a refinement of McCrary, 2008) to the running variable `margin_pres_party`. Under the null, the density is continuous at the cutoff $c = 0$. Manipulation around the threshold (e.g., recounts that systematically favor one side) would produce a discontinuity. Table A1 reports test statistics by sample.

Table A1: Density test of running variable at the cutoff

Sample	T	p	N
Full FABS sample	+1.11	0.269	2,123
Traditional regime	+0.78	0.437	700
Banned regime	+0.61	0.542	1,423

Notes. Cattaneo et al. (2018) jackknife-corrected local polynomial density test. Test statistic T has standard normal limiting distribution under the null of continuity at $c = 0$. None rejects continuity at any conventional level.

A.2 Pre-treatment covariate balance

We run the same RDD design with pre-treatment quantities as the dependent variable. If treatment status (aligned winner) is locally as-if random at the cutoff, pre-treatment quantities should be balanced (RDD coefficient ≈ 0). Table A2 reports the results.

B Robustness Specifications

This appendix reports the four robustness specifications summarized in Section 4.6, in greater detail.

Table A2: Pre-treatment covariate balance at the cutoff

Sample	Pre-treatment outcome	τ	SE	p	BW	N eff
Traditional	Prior-cycle margin	-0.018	(0.025)	0.493	0.078	159
Traditional	ln pre-period grants $[t-1, t]$	+2.773	(3.199)	0.263	0.100	216
Traditional	ln pre-pre grants $[t-3, t-2]$	+0.076	(2.232)	0.864	0.116	264
Banned	Prior-cycle margin	+0.040*	(0.025)	0.098	0.089	410
Banned	ln pre-period grants $[t-1, t]$	+0.415	(0.309)	0.146	0.171	908
Banned	ln pre-pre grants $[t-3, t-2]$	+0.395	(0.305)	0.156	0.140	719
Full	Prior-cycle margin	+0.022	(0.018)	0.181	0.091	603
Full	ln pre-period grants $[t-1, t]$	+0.685	(1.404)	0.687	0.096	660
Full	ln pre-pre grants $[t-3, t-2]$	-0.420	(1.865)	0.695	0.094	641

Notes. Local linear RDD with the same MSE-optimal bandwidth and triangular kernel as the main specifications, state-clustered SEs. The Banned-regime prior-cycle margin shows a borderline discontinuity ($p = 0.098$) consistent with mild persistence of partisan margins across House cycles. The $[t-1, t]$ pre-period grants placebo on the Banned sample is null at conventional thresholds and dissipates entirely under the FY-by-FY decomposition (Section 4.5). * $p < 0.10$.

B.1 FABS-only versus CFFR-extended panels

Table A3 compares headline specifications across the two data lineages (FABS-only versus CFFR-extended Traditional regime). The Banned regime is identical across panels because CFFR coverage ends in FY2010; the Traditional regime nearly doubles in sample size with no shift in point estimate or sign.

Table A3: Comparison: FABS-only vs. CFFR-extended panels

Specification	FABS panel	CFFR-extended panel	Verdict
<i>Per-regime estimates (FY$\geq$2007 unchanged)</i>			
Banned τ , level $[t+1, t+2]$	+0.530**, $N=1,423$	+0.530**, $N=1,423$	Stable
Banned τ , level $[t+2, t+3]$	+0.516*	+0.516*	Stable
Banned τ , delta $[t+1, t+2]$	+0.298**	+0.298**	Stable
<i>Traditional regime expansion</i>			
Traditional τ , level $[t+1, t+2]$	+0.069, $N=700$	+0.058, $N=1,383$	Strengthens null
Traditional τ , delta $[t+1, t+2]$	untestable	-0.275 null, $N=1,343$	Newly testable, opposite-signed from Banned
Traditional delta placebo	untestable	+0.113 null	Clean placebo
<i>Formal regime-interaction tests</i>			
Trad $\times$ Banned, level $h=0.15$ linear	$\Delta\tau = +0.624, p=0.083$	$\Delta\tau = +0.530, p=0.309$	FABS p partly artifact
Trad $\times$ Banned, delta $h=0.05$ linear	untestable	$\Delta\tau = +1.139, p=0.028^{**}$	First spec at $p < 0.05$
Trad $\times$ Banned, delta $h=0.075$ quadratic	untestable	$\Delta\tau = +1.150, p=0.074^*$	Marginal

Notes. Across 12 delta-DV specifications (bandwidth $\times$ polynomial order), all are sign-stable positive; only the narrowest-bandwidth specification crosses $p < 0.05$, so multiple-testing concerns apply. The level-DV regime interaction's borderline $p = 0.083$ on the FABS-only panel reflects in part a precision artifact in the FABS pre-2007 zero-coverage. * $p < 0.10$, ** $p < 0.05$.

B.2 FY-by-FY decomposition (full grid)

Table A4 reports the full FY-by-FY decomposition that backs Figure 3. Each cell is a separate RDD coefficient with state-clustered SEs.

Table A4: FY-by-FY RDD decomposition

Sample	Pure pre-period			Treatment-era		
	FY $t-2$	FY $t-1$	FY t	FY $t+1$ (hybrid)	FY $t+2$	FY $t+3$
Banned (2010, 14, 16, 18)	+0.264 ($p=0.57$)	+0.259 ($p=0.41$)	+0.275 ($p=0.40$)	+0.567* ($p=0.07$)	+0.499* ($p=0.09$)	+0.476* ($p=0.08$)
Banned, excl. 2010	+0.544 ($p=0.18$)	+0.359 ($p=0.19$)	+0.363 ($p=0.36$)	+0.774** ($p=0.03$)	+0.792*** ($p<0.01$)	+0.660** ($p=0.02$)
Traditional (2006, 2008)	+0.264 ($p=0.73$)	-0.107 ($p=0.87$)	+0.088 ($p=0.91$)	+0.027 ($p=0.70$)	+0.078 ($p=0.61$)	-0.164 ($p=0.68$)

Notes. Local linear RDD on $\log(\text{discretionary grants}+1)$ at fiscal-year resolution. Banned-regime pre-period FYs are all null; treatment-era FYs are uniformly positive at conventional levels. Traditional regime is null throughout. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 RDD specification stability

Coefficients are stable across bandwidth (MSE-optimal vs. uniform 5pt), kernel (triangular vs. uniform), and polynomial order (local linear vs. local quadratic) choices. The MSE-optimal triangular-kernel local-linear specification is the primary reference reported in the main text.

C State-Pres-Swing Threshold Sensitivity (Banned Regime)

Banned-regime close-election RDD on $\log$ discretionary grants stratified by state-preswingness, varying the swing/lean/safe threshold definitions. Table A5 reports the safe-state row across five threshold pairs; the (5pt, 10pt) row reproduces Section 4.3's main-text result. The safe-state τ is positive and statistically significant at the 5% level across all five threshold pairs (one of five at the 1% level); point estimate range 0.80–1.36, with

the main-text (5pt, 10pt) value at 1.09 sitting in the middle. The swing-state coefficient is essentially null and the lean-state coefficient is small in every parameterization (full table in `output/rd_kr_state_swing_sensitivity.csv`). The safe-state concentration is therefore not a knife-edge artifact of the (5pt, 10pt) cutpoints.

Table A5: Banned-regime RDD safe-state τ across state-pres-swing threshold pairs

Threshold (swing, safe)	τ	SE robust	p	BW	N total	N eff
(3pt, 7pt)	+0.799**	0.360	0.011	0.083	1,003	252
(5pt, 10pt) — main text	+1.092**	0.490	0.011	0.078	831	177
(5pt, 15pt)	+1.193**	0.667	0.030	0.080	664	137
(7pt, 12pt)	+1.358***	0.545	0.004	0.073	751	148
(10pt, 15pt)	+1.193**	0.667	0.030	0.080	664	137

Notes. Safe-state row from the Banned-regime stratified RDD across five swing/lean/safe threshold pairs. Local linear RDD with MSE-optimal bandwidth, triangular kernel, state-clustered SEs. Point estimate range 0.80–1.36; safe-state concentration is robust to threshold variation. The safe-state row depends only on the safe threshold, so threshold pairs sharing a safe cutoff—(5pt, 15pt) and (10pt, 15pt)—produce identical rows mechanically. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D FPDS Procurement Scope Robustness

The substitution argument predicts that the executive-channel signature should appear in grant categories where executive discretion is least constrained by statutory or procurement-process rules. Federal procurement (FPDS), governed by the Federal Acquisition Regulation, competitive bidding requirements, and cost/quality scoring standards, is a category where executive discretion is constrained much more tightly than in FABS discretionary grants. The substitution mechanism therefore predicts that the close-election RDD on FPDS procurement obligations should yield null or attenuated effects in both regimes, unlike the FABS-grants case where the Banned regime shows a sharp positive coefficient at the close-election margin.

Table A6 reports the close-election RDD on $\log(\text{post procurement obligations} + 1)$ aggregated by district-fiscal-year, separately within each regime and window. All FPDS coefficients are null at conventional thresholds. The Banned-regime FPDS coefficient un-

der the primary $[t+2, t+3]$ window is $\tau = -0.021$ ($p = 0.961$); under the $[t+1, t+2]$ window it is $\tau = -0.019$ ($p = 0.974$). Excluding the 2010 cycle yields $\tau = +0.070$ ($p = 0.747$). The contrast with the FABS-grants Banned-regime estimate ($\tau = +0.530$, $p = 0.034$ under the $[t+1, t+2]$ window) is sharp: the executive-channel signature is absent in procurement, exactly as the substitution mechanism’s scope condition predicts.

Table A6: FPDS procurement close-election RDD by regime

Window	Regime	τ	SE robust	p	BW	N total	N eff
$[t+1, t+2]$	Traditional	-0.301	0.318	0.266	0.151	700	374
$[t+1, t+2]$	Banned	-0.019	0.340	0.974	0.082	1,423	384
$[t+1, t+2]$	Banned excl. 2010	+0.070	0.402	0.747	0.076	1,044	229
$[t+1, t+2]$	Transition (2020)	-0.501	0.485	0.310	0.080	381	107
$[t+1, t+2]$	Restored (2024)	+0.514	0.647	0.323	0.124	377	177
$[t+2, t+3]$	Traditional	-0.244	0.327	0.333	0.144	700	355
$[t+2, t+3]$	Banned	-0.021	0.338	0.961	0.084	1,423	394
$[t+2, t+3]$	Banned excl. 2010	-0.031	0.400	0.976	0.079	1,044	241
$[t+2, t+3]$	Transition (2020)	-0.250	0.544	0.816	0.080	381	107
Placebo, $[t+1, t+2]$	Banned, pre-period	+0.150	0.345	0.553	0.090	1,423	421

Notes. Close-election RDD on $\log(\text{post procurement obligations} + 1)$ aggregated by district-fiscal-year using FPDS contract types A, B, C, D. Local linear RDD with MSE-optimal bandwidth, triangular kernel, state-clustered SEs. None of the regime-specific procurement RDDs reach conventional significance; all coefficients are null. Compare to FABS grants Banned $\tau = +0.530$ ($p = 0.034$) under the same window. The procurement category is governed by the Federal Acquisition Regulation and competitive-bidding rules that bind executive discretion much more tightly than FABS grants; the null procurement coefficients are consistent with the substitution mechanism’s scope condition that the executive-channel alignment effect should appear in grant categories where executive discretion is least constrained, and not in procurement.

The procurement-null result is itself an indirect identification check on the FABS-grants Banned-regime positive estimate. If the FABS-grants positive were generated by a generic “winning party gets more federal dollars” mechanism that applied across all federal-spending categories, the same close-election RDD on procurement would also yield positive coefficients. The procurement null rejects this generic interpretation and supports the more specific substitution-mechanism reading: executive discretion over discretionary grants is the operative channel, and the close-election alignment effect emerges precisely where executive discretion is institutionally available.

E Heterogeneous TWFE Conditional Alignment Effects (Joint Tests)

For each safety stratum and each regime, Table A7 reports the conditional alignment effect (the linear combination of the main `aligned_winner` coefficient and the alignment-by-stratum interaction from the district-FE TWFE specification of Section 4.2) with joint SE and 95% CI computed via the post-estimation `lincom` on the clustered `vcov`. Reference category: safety = safe.

Table A7: Conditional alignment effects within each safety stratum (joint tests)

Regime	Safety stratum	n post-singleton	Conditional β	Joint SE	95% CI	p
Traditional	safe	505	+0.171	0.052	[+0.069, +0.273]	0.001
Traditional	moderate	102	−0.181	0.115	[−0.407, +0.045]	0.117
Traditional	competitive	75	+0.112	0.148	[−0.178, +0.402]	0.448
Banned	safe	965	+0.055	0.056	[−0.054, +0.165]	0.323
Banned	moderate	262	−0.153	0.219	[−0.582, +0.277]	0.486
Banned	competitive	175	−0.196**	0.085	[−0.363, −0.029]	0.021

Notes. Specification: $\log(\text{post_assistance} + 1) \sim \text{aligned_winner} \times \text{safety} \mid \text{district FE} + \text{year FE}$, cluster $\sim$ state-by-district. Reference: safety = safe. Conditional alignment effect within stratum s equals the linear combination of the main `aligned_winner` coefficient and the corresponding interaction; joint SE from $c'Vc$. Traditional regime concentrates positive alignment in safe seats. Banned regime shows no significant safe-seat concentration; only the competitive stratum is significant, and it is negative. Reconciliation with the close-election RDD is discussed in Section 4.2.

F Signed-Stratum Robustness

This appendix reports robustness diagnostics for the signed-stratum split of the Banned-regime close-election RDD reported in Section 4.3. The opposition-stronghold cell estimates $\tau = +1.616$ on a small effective sample ($N_{\text{eff}} = 39$) at an MSE-optimal bandwidth of $h = 0.048$. Because the signed split partitions the Banned close-election sample into five strata, the opposition-stronghold result also survives a family-wise multiple-comparison correction: its conventional MSE-optimal $p = 3.9 \times 10^{-5}$ scales to a Bonferroni-adjusted

$\approx 2 \times 10^{-4}$ across the five strata, and the bias-corrected estimate is significant at the one-percent level or better at every fixed bandwidth in Table A8, so a five-stratum Bonferroni correction leaves it significant throughout. Three diagnostics establish that the coefficient is not driven by bandwidth choice or single-state outliers: a fixed-bandwidth scan, a leave-one-state-out re-estimation, and a tabulation of the in-bandwidth observation set.

F.1 Fixed-bandwidth scan

Table A8 reports the close-election RDD for the safe-opposition and safe-co-partisan cells of the Banned regime across fixed bandwidths from $h = 0.05$ to $h = 0.20$, pairing the conventional local-linear estimate with its conventional p -value and the bias-corrected estimate with its robust p -value. The conventional safe-opposition coefficient remains in the +1.4 to +2.0 range across narrow-to-moderate bandwidths ($h \leq 0.10$), significant at the one-percent level throughout, and attenuates toward zero at $h = 0.15$ and $h = 0.20$ as districts well outside the close-election neighborhood dominate the local fit. The bias-corrected estimate corrects precisely this wide-bandwidth smoothing bias and remains in the +1.6 to +2.3 range, statistically significant at all five bandwidths. The safe-co-partisan coefficient is stable across all bandwidths in the +0.65 to +0.95 range, reaching conventional significance only as the bandwidth widens ($h \geq 0.15$). The operative bandwidths for close-election identification are below $h = 0.15$; within that range the safe-opposition coefficient is large, stable, and robust under both inference modes.

F.2 Leave-one-state-out

Table A9 reports the safe-opposition coefficient under each leave-one-state-out re-estimation. The coefficient ranges from +1.437 (dropping Texas) to +2.183 (dropping California) across the 32 states that contribute to the cell. No single state moves the estimate by more than 0.6 in absolute terms; California is the largest mover and pushes

Table A8: Fixed-bandwidth RDD for safe substrata (Banned regime)

h	Safe, opposition stronghold				Safe, co-partisan stronghold			
	τ_{conv}	p_{conv}	τ_{bc}	p_{rob}	τ_{conv}	p_{conv}	τ_{bc}	p_{rob}
0.050	+1.725	<0.001	+1.637	0.001	+0.648	0.305	+0.752	0.388
0.075	+1.955	<0.001	+2.090	<0.001	+0.864	0.088	+0.506	0.481
0.100	+1.440	<0.001	+2.254	0.002	+0.890	0.056	+0.808	0.191
0.150	+0.531	0.411	+2.194	<0.001	+0.948	0.010	+0.844	0.109
0.200	-0.027	0.967	+1.674	<0.001	+0.890	0.005	+0.995	0.031

Notes. Local linear RDD at fixed bandwidth h , triangular kernel, state-clustered SEs. $\tau_{\text{conv}}/p_{\text{conv}}$ pair the conventional local-linear estimate with its conventional p -value; $\tau_{\text{bc}}/p_{\text{rob}}$ pair the bias-corrected estimate with its robust p -value (Calonico et al., 2014). The conventional safe-opposition estimate attenuates at wide bandwidths as observations far from the cutoff dominate the local fit; the bias-corrected estimate, which corrects this wide-bandwidth smoothing bias, remains in the +1.6 to +2.3 range and statistically significant at all five bandwidths.

the estimate down rather than up, indicating that the +1.616 baseline is conservative relative to the without-California estimate.

Table A9: Leave-one-state-out re-estimates, safe-opposition cell (Banned)

Dropped state	τ (LOO)	Shift from baseline (+1.616)
California (CA)	+2.183	+0.567
Texas (TX)	+1.437	-0.179
New Jersey (NJ)	+1.498	-0.118
Alabama (AL)	+1.670	+0.054
Arkansas (AR)	+1.575	-0.041
Kentucky (KY)	+1.656	+0.040
Illinois (IL)	+1.577	-0.039
Other 25 states	$\in [+1.604, +1.619]$	$ \text{shift} < 0.013$

Notes. Local linear RDD on the safe-opposition cell of the Banned regime, MSE-optimal bandwidth, triangular kernel, state-clustered SEs. Seven states with the largest LOO movement are listed individually; the remaining 25 contributing states all leave the estimate within ± 0.013 of the baseline.

F.3 In-bandwidth observation set

Within the MSE-optimal bandwidth ($h = 0.048$), the safe-opposition cell contains 40 district-cycle observations spanning 12 states across the four Banned-regime election cycles (2010, 2014, 2016, 2018). Two empirical groups dominate. The first comprises 2016- and 2018-cycle Democratic close wins in deep-Democratic states under the Trump admin-

istration: Hill (CA-25), Rouda (CA-48), Harder (CA-10), Porter (CA-45), Cisneros (CA-39), Cox (CA-21), Carbajal (CA-24), Bera (CA-7), Kim (NJ-3), Malinowski (NJ-7), Van Drew (NJ-2), Casten (IL-6), Underwood (IL-14), Schneider (IL-10), Schrier (WA-8). The second comprises Republican close wins in deep-Democratic states (Issa, Hunter, Nunes, McClintock, Knight, Denham, McMorris-Rodgers, Herrera-Beutler) and Democratic close wins in deep-Republican states (Matheson UT-2, Doggett TX-25, Chandler KY-6, Ashford NE-2). The cell is geographically dispersed across the 12 states; no two-or-three-district concentration accounts for the coefficient.